

Notes: Yogo, Whitten, and Cox (JFE, 2025)

Financial Inclusion Across the United States

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1 Big picture

- **Question.** How unequal is financial participation across the United States, and what explains low participation in retirement and bank accounts among low-income households?
- **Main setting.** The paper studies the universe of U.S. households with at least one member aged 50 to 59 in administrative tax data from 2015 to 2019. This population is in prime retirement-saving years and is large enough to study very fine geographic variation.
- **Main outcomes.** The two primary outcomes are retirement account participation and bank account participation. Retirement accounts include employer retirement plans, both defined benefit and defined contribution plans, and individual retirement arrangements (IRAs). Bank account participation is proxied using electronic funds transfer information on tax returns and interest information returns.
- **Core descriptive fact.** Financial participation is much lower for households in the lowest income quintile than for richer households. In the lowest income quintile in 2019, roughly 21 percent have retirement accounts. Bank account participation is also lower for low-income households, though the level differs depending on whether it is measured in tax data or survey data.
- **Geographic fact.** Even conditional on income, financial participation varies greatly across ZIP Code Tabulation Areas (ZCTAs). Some low-income households participate at high rates, while others in similarly low-income brackets do not.
- **Retirement-account geography.** Among low-income households, retirement participation is positively related to neighborhood average income and economic connectedness, and negatively related to neighborhood income inequality and minority population shares, even after controlling for household income and regional price parities.
- **Bank-account geography.** Bank participation among low-income households is less strongly connected to racial composition but is negatively related to neighborhood income inequality. The presence of a bank branch has a small positive association.
- **Causal result.** Access to an employer retirement plan is a major determinant of retirement account participation. The paper uses a panel IV design based on whether the worker's 2010 employer later starts offering a retirement plan.
- **Policy result.** Recent proposals or state policies for universal access to retirement plans with automatic enrollment could substantially raise retirement account participation among low-income households. The paper estimates a 19 percentage point increase in the lowest income quintile over ten years.
- **Economic takeaway.** Low financial participation is not only an income problem. Geography, peer environment, access to employer retirement plans, and institutional design all matter. Financial inclusion policy should therefore target both access and take-up frictions.

2 Data construction

2.1 Administrative tax data

- The administrative tax data come from the Internal Revenue Service and cover tax years 2006 to 2019.
- The data contain Form 1040 tax returns and many information returns, including W-2, 1099-INT, 1099-DIV, 1099-R, 1099-MISC, 1099-G, SSA-1099, 1099-B, Form 5498, and Form 1095.
- The main analysis sample consists of all individuals aged 50 to 59 on April 1 of each year from 2015 to 2019 who have either a tax return or an information return with a ZIP code in U.S. states or Washington, DC.
- The authors link individuals into households using tax filing status, shared tax returns, household identifiers, and Form 1095 links.
- For married households, the household is treated as participating in an account if either spouse participates.

Interpretation: The administrative data avoid the small-sample and measurement-error problems of household surveys. They are especially valuable for mapping financial participation at the ZCTA level.

2.2 Why the age 50–59 sample matters

- Households aged 50 to 59 are close enough to retirement that retirement account participation is economically important.
- This group is young enough that most households are not yet drawing retirement benefits, but old enough that access to retirement plans has had time to matter.
- Studying this age group helps separate financial inclusion from very young adult bank access, which has been a more common focus of financial inclusion research.

Interpretation: The paper reframes financial inclusion in the United States as a retirement-saving and asset-accumulation issue, not only as a bank-account access issue.

2.3 Household income construction

- The paper constructs pre-tax household income following Larrimore et al. (2021).
- For tax filers, the starting point is total income on Form 1040.
- The authors adjust total income by adding tax-exempt interest, replacing taxable private retirement income with gross private retirement income from Form 1099-R, and replacing taxable Social Security benefits with total Social Security benefits from SSA-1099.
- For nonfilers, income is constructed from information returns.
- Pre-tax household income is truncated at zero to limit the effect of business losses.

Interpretation: Income is constructed to approximate household resources rather than just taxable income. This is important because retirement and Social Security income can be partly nontaxable.

2.4 Usual income and income quintiles

- The paper ranks households by usual income rather than by a single-year income realization.
- Usual income is designed to reduce noise from temporary earnings fluctuations, capital gains, business losses, or one-time shocks.

- Households are grouped into usual-income quintiles, and most tables report participation by the 0–20, 20–40, 40–60, 60–80, and 80–100 percentiles.

Interpretation: Using usual income makes the financial participation patterns closer to long-run economic status rather than transitory income.

2.5 Retirement account participation

The main retirement participation variable is:

$$RetirementAccount_{h,t} = 1 \quad (1)$$

if household h has an open retirement account in year t .

- Retirement accounts include employer retirement plans and IRAs.
- Employer retirement plans include both defined benefit and defined contribution plans.
- The paper uses a ten-year history to identify whether an individual has an account that is still open.
- A retirement account is identified as closed when Form 1099-R reports total distribution, excluding rollovers and recharacterized IRA contributions.

Interpretation: The ten-year history matters because a person might not receive a retirement information return every year even if an account remains open.

2.6 Access to employer retirement plans

The paper defines access to an employer retirement plan using Form W-2:

$$EmployerPlanAccess_{i,t} = 1 \quad (2)$$

if the retirement-plan box is checked on any Form W-2 issued by individual i 's employer.

- The authors search all W-2s, not only the W-2s of sampled individuals, to infer whether an employer offers a plan.
- The individual's income from that employer must exceed the federal minimum wage times 1000 hours to infer benefit eligibility.
- For households, access is defined as access by either spouse.
- The paper also identifies whether an employer plan has automatic enrollment by linking to Form 5500 filings.

Interpretation: Access to an employer plan is central because most retirement account participation among low-income households comes through employer plans rather than stand-alone IRAs.

2.7 Bank account participation

The paper measures bank account participation using two tax-data indicators:

- electronic funds transfer for payment of taxes or receipt of refunds on Form 1040;
- taxable interest income from Form 1099-INT.

The benchmark tax-data measure uses a ten-year history:

$$BankAccount_{h,t} = 1 \quad (3)$$

if household h shows evidence of bank account participation in year t or the previous nine years.

Interpretation: This measure is imperfect because a household could have a bank account without using electronic tax payments or earning reportable interest. Therefore the paper compares it with the Survey of Consumer Finances and reports an alternative measure in the appendix.

2.8 Survey of Consumer Finances comparison

- The Survey of Consumer Finances (SCF) provides a survey benchmark for retirement and bank account participation.
- The SCF is smaller but directly asks about financial accounts.
- The tax data and SCF align closely for retirement accounts, but bank account participation differs because the tax-data proxy captures tax-related electronic account use and interest reporting.

Interpretation: The administrative data offer granularity and population coverage, while the SCF offers direct account measures. The paper uses both to understand measurement error.

2.9 ZCTA-level geographic variables

The paper studies whether local environment explains participation among low-income households. The ZCTA-level variables are:

- average log income;
- standard deviation of log income, interpreted as local income inequality;
- racial composition: Hispanic, Black, Asian, and other nonwhite population shares;
- economic connectedness, based on Chetty et al. (2022);
- bank branch access and bank branch density;
- regional price parities, used to control for geographic cost-of-living differences.

Interpretation: These variables allow the paper to test whether financial participation is shaped by peer effects, local opportunity, racial composition, and supply-side financial access.

3 Descriptive facts by income

3.1 Retirement account participation

- Table 1 shows that retirement account participation rises sharply with usual income.
- In the administrative tax data, the lowest income quintile has about 21–22 percent retirement account participation from 2015 to 2019.
- The highest income quintile has about 97 percent participation.
- The SCF provides broadly similar retirement participation rates, validating the administrative-tax-data measure.

Interpretation: Retirement account participation is extremely unequal by income. The highest quintile is almost universally covered, while most low-income households lack retirement accounts.

3.2 Bank account participation

- Table 2 shows that bank account participation also rises with income.
- In the tax data, the lowest income quintile has about 67–71 percent bank account participation from 2015 to 2019.
- In the SCF, the lowest income quintile has higher reported bank account participation: 76 percent in 2016 and 80 percent in 2019.
- Higher-income households have participation rates close to one.

Interpretation: Bank participation is more common than retirement participation, but there remains a substantial inclusion gap at the bottom of the income distribution.

3.3 Composition of retirement account participation

- Table 3 decomposes retirement participation into employer-plan-only, IRA-only, and both.
- In the lowest income quintile, most retirement participants use employer plans rather than IRAs only.
- The IRA-only category is small across all income groups.
- Higher-income households are much more likely to have both an employer plan and an IRA.

Interpretation: The importance of employer plans motivates the causal analysis of employer retirement plan access.

3.4 Access to employer retirement plans

- Table 4 shows that access to employer retirement plans also rises sharply with income.
- In the lowest income quintile, access is about 38–42 percent across 2015–2019.
- In the highest income quintile, access is around 96–97 percent.

Interpretation: The retirement participation gap is partly an access gap: low-income households are much less likely to work for employers that offer retirement plans.

4 Geography and financial participation

4.1 Map evidence

- Figures 1 and 2 map financial participation by ZCTA for households in the lowest income quintile.
- Figure 1 maps retirement account participation.
- Figure 2 maps bank account participation.
- Both maps show substantial within-region variation: low-participation and high-participation ZCTAs are often geographically close.

Interpretation: The maps show that low participation is not simply a national, regional, or state-level phenomenon. Local environment matters.

4.2 Retirement account participation regression

The main geographic regression for retirement participation is:

$$Retire_{h,z} = \alpha + X'_h\beta + Z'_z\gamma + \varepsilon_{h,z}, \quad (4)$$

where X_h includes household characteristics and Z_z includes ZCTA-level characteristics.

- The sample is households in the lowest income quintile with a member aged 50 to 59 in 2019.
- Household controls include log income, marital status, and regional price parities.
- ZCTA controls include average log income, inequality, racial composition, economic connectedness, and employer-plan access.

Interpretation: The regression asks whether local characteristics predict financial participation after holding household income and cost of living fixed.

4.3 Retirement geography results

- A standard deviation increase in household log income predicts about a 6 percentage point increase in retirement participation in the benchmark specification.
- Average ZCTA income is positively related to retirement participation.
- ZCTA income inequality is negatively related to retirement participation.
- Hispanic and Black population shares are negatively related to retirement participation.
- Economic connectedness is positively related to retirement participation.
- Access to an employer retirement plan is a very strong predictor of retirement participation.
- Having a bank account is also positively related to having a retirement account, suggesting a common participation factor such as trust in financial institutions.

Interpretation: The local social and economic environment matters even conditional on household income. Peer effects, local inequality, and employer benefits all appear important.

4.4 Bank account participation regression

The bank-participation regression is analogous:

$$Bank_{h,z} = \alpha + X'_h\beta + Z'_z\gamma + \varepsilon_{h,z}. \quad (5)$$

- The sample is restricted to households with tax filings in the previous ten years, where bank account measurement is more reliable.
- The model includes household income, marital status, regional price parities, ZCTA income measures, racial composition, economic connectedness, and bank branch measures.

Interpretation: The comparison between retirement and bank participation helps distinguish plan-access mechanisms from broader financial inclusion mechanisms.

4.5 Bank geography results

- Household log income is positively related to bank account participation.
- ZCTA income inequality is negatively related to bank account participation.
- Average ZCTA income is much less important for bank accounts than for retirement accounts.
- Minority population shares generally do not predict bank participation in the same way they predict retirement participation, except for the other-nonwhite category.
- Economic connectedness is not a strong predictor of bank account participation.
- The presence of a bank branch is positively related to bank participation, but branch density is not.

Interpretation: Retirement participation seems more connected to employer access and local social environment, while bank participation has different and more limited geographic determinants.

4.6 Open interpretations

- Income matters directly because households without enough income cannot save and may face fees on low-balance accounts.
- Financial literacy and education may matter because low-income households often have lower financial literacy.
- Peer effects may matter because households learn about financial products from neighbors, coworkers, and social networks.
- Racial composition may capture unobserved household race, discrimination, labor-market opportunities, or mistrust of financial institutions.
- The paper cannot fully distinguish household-level race from neighborhood-level racial composition because the tax data do not contain race.

Interpretation: The geographic results are descriptive but policy-relevant because they identify where financial inclusion gaps are concentrated.

5 Employer retirement plans and causal identification

5.1 Endogeneity problem

Access to an employer retirement plan is not randomly assigned. The structural relation of interest is:

$$Retire_{h,2019} = \alpha + \beta_1 Access_h + \beta_2 AddYears_h + \beta_3 AutoEnroll_h + X'_h \delta + \varepsilon_h. \quad (6)$$

- $Access_h$ is an indicator for access to an employer retirement plan by either spouse from 2011 to 2019.
- $AddYears_h$ is additional years beyond the first of access without automatic enrollment.
- $AutoEnroll_h$ is an indicator for automatic enrollment in an employer retirement plan.
- The outcome is retirement account participation in 2019.

Interpretation: Naive OLS would be biased because workers who care about retirement saving may choose employers with retirement plans.

5.2 Instrument construction

The instrument is based on the worker's employer in 2010. The idea is to ask what plan access the worker would have received had she stayed with the same employer from 2010 to 2019.

- The sample includes households with a member aged 50 to 59 in 2019 who lacked access to an employer plan in 2010.
- The intent-to-treat instrument is the counterfactual access to an employer retirement plan if the worker remained with the 2010 employer.
- The instrument has three components corresponding to the three endogenous variables: access, additional years of access, and automatic enrollment.

Interpretation: The design uses unexpected employer plan adoption after 2010 as a source of variation in plan access.

5.3 Illustration of the instrument

- In Panel A of Fig. 3, a worker stays with employer A, and employer A starts a retirement plan in 2016.
- In Panel B, the worker switches from employer A to employer B, and employer B starts a retirement plan in 2013.
- Realized access differs from counterfactual access because the worker changes employers.
- The instrument uses the counterfactual path based on the 2010 employer rather than the realized employer path.

Interpretation: The identifying assumption is that whether the 2010 employer later starts a retirement plan is not predictable from the worker's unobserved saving preferences, conditional on employer size, industry, and worker controls.

5.4 Threats to identification

- Workers with stronger saving motives might choose employers more likely to start retirement plans.
- More productive workers could help firms grow, and growing firms may start retirement plans.
- Workers with existing retirement security may choose employers without retirement benefits.

- The paper controls for income, employer size, industry, marital status, and other characteristics, but residual productivity or preference differences could remain.

Interpretation: The IV design is credible because it uses employer changes over time, but it still relies on a conditional exogeneity assumption.

5.5 Instrumental variables specification

The first-stage system can be written as:

$$Access_h = \pi_{10} + \pi_{11}Z_h^{Access} + \pi_{12}Z_h^{Years} + \pi_{13}Z_h^{Auto} + X_h'\rho_1 + u_{1h}, \quad (7)$$

$$AddYears_h = \pi_{20} + \pi_{21}Z_h^{Access} + \pi_{22}Z_h^{Years} + \pi_{23}Z_h^{Auto} + X_h'\rho_2 + u_{2h}, \quad (8)$$

$$AutoEnroll_h = \pi_{30} + \pi_{31}Z_h^{Access} + \pi_{32}Z_h^{Years} + \pi_{33}Z_h^{Auto} + X_h'\rho_3 + u_{3h}. \quad (9)$$

The second stage is:

$$Retire_{h,2019} = \alpha + \beta_1\widehat{Access}_h + \beta_2\widehat{AddYears}_h + \beta_3\widehat{AutoEnroll}_h + X_h'\delta + \varepsilon_h. \quad (10)$$

Interpretation: The model allows the first year of access, additional years of access, and automatic enrollment to have different marginal effects.

5.6 Causal effects

- Table 7 shows that access to an employer retirement plan substantially raises retirement account participation.
- In the lowest income quintile, access raises participation by about 32 percentage points.
- Automatic enrollment adds about 28 percentage points in the lowest income quintile.
- Each additional year of access beyond the first without automatic enrollment has a smaller incremental effect, about one percentage point in the lowest quintile.
- The effects are heterogeneous across income groups and generally smaller in the highest quintile because baseline participation is already high.

Interpretation: Employer access and automatic enrollment are powerful tools for financial inclusion among low-income households.

5.7 Universal access counterfactual

The policy counterfactual asks what would happen if all workers had access to an employer retirement plan with automatic enrollment from 2010 to 2019.

$$\Delta Participation_q = \frac{1}{N_q} \sum_{h \in q} \left[\widehat{P}_h^{Universal} - \widehat{P}_h^{Observed} \right]. \quad (11)$$

- The prediction uses the IV coefficients from Table 7.
- The counterfactual access variable equals one for a worker-year if the worker received a W-2.
- Household access is one if either spouse works and therefore would have access.
- Automatic enrollment is imposed in the counterfactual.

Interpretation: This is a policy simulation, not an estimate from an already implemented national policy. External validity depends on whether mandated access behaves like voluntary employer plan adoption.

5.8 State-sponsored retirement savings programs

- Ten states had mandates requiring many employers to enroll workers in state-sponsored retirement savings programs if they did not already offer a plan.
- These programs are legally IRAs and are subject to IRA contribution limits.
- Default contribution rates are typically 3 or 5 percent.
- Workers can adjust the contribution rate or opt out.
- Because these mandates are recent, the paper cannot yet observe their long-run effect.

Interpretation: State auto-IRA programs motivate the counterfactual analysis but are too new to evaluate fully in the paper's data.

6 Tables and figures

6.1 Tables 1–4: baseline financial participation by income

Read:

- Table 1 shows retirement account participation by income. Participation is about 21 percent for the lowest income quintile in 2019 and about 97 percent for the highest income quintile.
- Table 2 shows bank account participation by income. The tax-data measure is lower than the SCF measure in the bottom quintile, but both measures show a strong income gradient.
- Table 3 decomposes retirement accounts into employer-plan-only, IRA-only, and both. Employer plans are the dominant channel for low-income households.
- Table 4 shows access to employer retirement plans. The lowest income quintile has much lower access than higher-income groups.

Economic message: These tables establish the basic inclusion gap: low-income households are much less likely to hold retirement accounts and less likely to have employer-plan access.

Table 1

Retirement account participation by income.

Year	Percentile of usual income					All
	0–20	20–40	40–60	60–80	80–100	
Panel A. Administrative tax data						
2015	0.22	0.56	0.83	0.94	0.97	0.70
2016	0.21	0.55	0.82	0.94	0.97	0.70
2017	0.21	0.55	0.82	0.94	0.97	0.70
2018	0.21	0.55	0.81	0.93	0.97	0.69
2019	0.21	0.54	0.81	0.93	0.97	0.69
Panel B. Survey of Consumer Finances						
2016	0.33	0.62	0.83	0.90	0.95	0.72
2019	0.24	0.63	0.76	0.92	0.93	0.69

This table reports retirement account participation for households with a member aged 50 to 59. Retirement accounts include employer retirement plans (i.e., defined benefit and defined contribution plans) and IRAs.

Table 1: retirement account participation

Table 3

Breakdown of retirement account participation.

Households with	Percentile of usual income					All
	0–20	20–40	40–60	60–80	80–100	
Employer plan only	0.11	0.35	0.47	0.41	0.23	0.32
IRA only	0.05	0.06	0.06	0.05	0.05	0.05
Both	0.04	0.13	0.28	0.47	0.69	0.32
Total	0.21	0.54	0.81	0.93	0.97	0.69

This table reports a breakdown of retirement account participation for households with a member aged 50 to 59 in the 2019 administrative tax data.

measurement assumptions mimic the survey data. Several factors could explain why retirement account participation in the administrative tax data has a steeper income gradient. As we discussed in Section 2, the Survey of Consumer Finances uses a broader definition of income that includes food stamps and other government support, which could lead to a different correlation with retirement account participation. Alternatively, because the Survey of Consumer Finances is a survey, both retirement account participation and income are subject to measurement or imputation error when households misreport or refuse to answer survey questions. Random measurement or imputation error in income attenuates the true relation between retirement account participation and income, which could explain the flatter income gradient in the Survey of Consumer Finances.

3.2. Bank account participation

Table 3: composition of retirement participation

Table 2

Bank account participation by income.

Year	Percentile of usual income					All
	0–20	20–40	40–60	60–80	80–100	
Panel A. Administrative tax data						
2015	0.71	0.94	0.99	1.00	1.00	0.93
2016	0.70	0.94	0.99	1.00	1.00	0.92
2017	0.68	0.94	0.98	1.00	1.00	0.92
2018	0.67	0.93	0.98	0.99	1.00	0.92
2019	0.70	0.93	0.98	0.99	1.00	0.92
Panel B. Survey of Consumer Finances						
2016	0.76	0.93	0.98	0.99	1.00	0.93
2019	0.80	0.95	1.00	1.00	1.00	0.95

This table reports bank account participation for households with a member aged 50

Table 2: bank account participation

Table 4

Access to an employer retirement plan.

Year	Percentile of usual income					All
	0–20	20–40	40–60	60–80	80–100	
2015	0.42	0.81	0.93	0.97	0.96	0.82
2016	0.41	0.80	0.93	0.97	0.96	0.81
2017	0.39	0.80	0.93	0.97	0.96	0.81
2018	0.38	0.80	0.93	0.97	0.96	0.81
2019	0.38	0.79	0.93	0.96	0.96	0.80

This table reports access to an employer retirement plan (i.e., defined benefit and defined contribution plans) for households with a member aged 50 to 59 in the administrative tax data.

Table 4: access to employer retirement plans

6.2 Tables 5 and 6: geography of participation

Read:

- Table 5 reports regressions for retirement account participation in the lowest income quintile.
- Household income and marital status matter, but ZCTA characteristics also matter.
- Average local income and economic connectedness are positively related to retirement participation.
- Local income inequality and minority population shares are negatively related to retirement participation.
- Access to an employer retirement plan is a very large predictor of retirement account participation.
- Table 6 reports analogous regressions for bank account participation.
- Income inequality is negatively related to bank account participation, while racial composition has a weaker role than in the retirement account regression.
- A bank branch in the ZCTA has a small positive association with bank account participation.

Economic message: Financial participation differs across neighborhoods even among similarly low-income households. Retirement participation is especially tied to employer access and local social/economic environment.

Table 5
Geography of retirement account participation in the lowest income quintile.

Regressor	(1)	(2)	(3)	(4)	(5)	(6)
Log income	0.055 (0.002)	0.039 (0.000)	0.041 (0.000)	0.041 (0.000)	-0.006 (0.000)	-0.008 (0.000)
Single	-0.133 (0.002)	-0.123 (0.002)	-0.124 (0.002)	-0.124 (0.002)	-0.083 (0.001)	-0.076 (0.001)
Regional price parities	0.001 (0.001)	0.003 (0.001)	0.001 (0.001)	0.009 (0.001)	0.012 (0.001)	0.012 (0.001)
Average log income		0.012 (0.001)	0.012 (0.001)	0.012 (0.001)	0.008 (0.001)	0.008 (0.001)
SD of log income		-0.020 (0.001)	-0.017 (0.001)	-0.016 (0.001)	-0.011 (0.001)	-0.010 (0.001)
Racial composition:						
Hispanic			-0.106 (0.004)	-0.089 (0.004)	-0.091 (0.002)	-0.095 (0.003)
Black			-0.033 (0.003)	-0.021 (0.003)	-0.041 (0.003)	-0.044 (0.003)
Asian			-0.056 (0.020)	-0.065 (0.019)	0.039 (0.012)	0.033 (0.013)
Other nonwhite			-0.032 (0.006)	-0.027 (0.006)	-0.081 (0.006)	-0.078 (0.006)
Economic connectedness				0.011 (0.001)		
Access to employer plan					0.406 (0.001)	0.396 (0.001)
Bank account						0.039 (0.001)
Constant	0.329 (0.003)	0.343 (0.002)	0.370 (0.002)	0.369 (0.002)	0.166 (0.001)	0.136 (0.002)
R ²	0.03	0.04	0.04	0.04	0.26	0.04
Observations	7,189,565	7,189,541	7,189,541	7,173,593	7,189,541	7,189,541

This table reports regressions of retirement account participation on household and ZCTA-level characteristics. The household characteristics are log income, marital status, access to an employer retirement plan by either spouse from 2011 to 2019, and bank account participation. The ZCTA-level characteristics are regional price parities, average log income, standard deviation of log income, racial composition, and economic connectedness. All coefficients on the continuous variables, except for racial composition, are standardized. Robust standard errors, clustered by ZCTA, are reported in parentheses. The sample includes households in the lowest income quintile with a member aged 50 to 59 in the 2019 administrative tax data.

Table 5: retirement participation geography

Table 6
Geography of bank account participation in the lowest income quintile.

Regressor	(1)	(2)	(3)	(4)	(5)
Log income	0.064 (0.001)	0.062 (0.001)	0.062 (0.001)	0.061 (0.001)	0.062 (0.001)
Single	-0.055 (0.001)	-0.053 (0.001)	-0.053 (0.001)	-0.053 (0.001)	-0.054 (0.001)
Regional price parities	-0.003 (0.001)	0.003 (0.001)	0.002 (0.001)	0.004 (0.001)	0.003 (0.001)
Average log income		-0.005 (0.002)	-0.004 (0.002)	-0.004 (0.002)	-0.004 (0.002)
SD of log income		-0.027 (0.002)	-0.028 (0.003)	-0.028 (0.003)	-0.027 (0.003)
Racial composition:					
Hispanic			0.006 (0.004)	-0.001 (0.004)	0.014 (0.004)
Black			0.032 (0.004)	0.027 (0.004)	0.039 (0.004)
Asian			0.042 (0.008)	0.046 (0.008)	0.039 (0.008)
Other nonwhite			-0.096 (0.008)	-0.098 (0.008)	-0.086 (0.008)
Economic connectedness				-0.005 (0.001)	
Bank branch presence					0.012 (0.003)
Bank branch density					0.005 (0.001)
Constant	0.923 (0.001)	0.928 (0.001)	0.925 (0.001)	0.925 (0.001)	0.914 (0.003)
R ²	0.04	0.04	0.04	0.04	0.04
Observations	5,361,090	5,361,074	5,361,074	5,348,662	5,361,074

This table reports regressions of bank account participation on household and ZCTA-level characteristics. The household characteristics are log income and marital status. The ZCTA-level characteristics are regional price parities, average log income, standard deviation of log income, racial composition, economic connectedness, and bank branch presence and density. All coefficients on the continuous variables, except for racial composition, are standardized. Robust standard errors, clustered by ZCTA, are reported in parentheses. The sample includes households in the lowest income quintile with a member aged 50 to 59 in the 2019 administrative tax data, who have filed tax returns in the previous ten years.

Table 6: bank participation geography

6.3 Figures 1 and 2: maps of financial participation

Read:

- Figure 1 maps retirement account participation for low-income households by ZCTA.
- Figure 2 maps bank account participation for low-income households by ZCTA.
- Darker shades indicate more populous ZCTAs.
- Both maps show large geographic variation and many local pockets of low participation.

Economic message: The maps motivate a geographic analysis: financial inclusion gaps are local, not just national or regional.

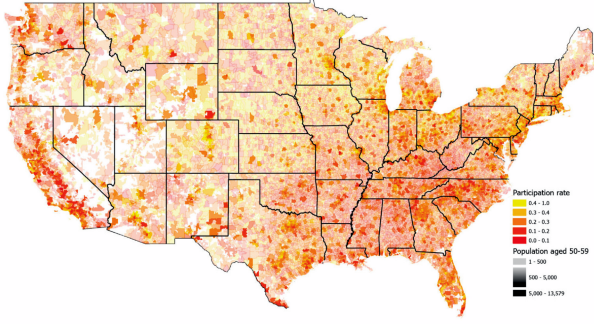


Figure 1: retirement account participation map

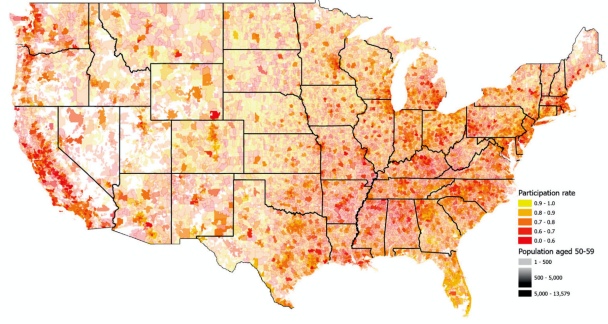


Figure 2: bank account participation map

6.4 Figure 3 and Table 7: IV strategy and causal effects

Read:

- Figure 3 illustrates the instrument based on counterfactual access had a worker remained with the 2010 employer.
- Table 7 reports the instrumental variables estimates.
- In the lowest income quintile, access to an employer plan raises retirement account participation by about 32 percentage points.
- Automatic enrollment raises participation further, by about 28 percentage points in the lowest quintile.
- Additional years of access have smaller but positive marginal effects.

Economic message: Employer retirement plan access is not merely correlated with participation; under the IV design, it has a large causal effect.

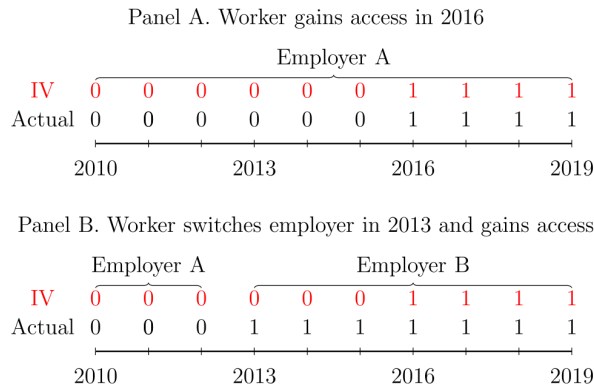


Figure 3: instrument illustration

Table 7

Instrumental variables regression for retirement account participation.

Regressor	Percentile of usual income				
	0-20	20-40	40-60	60-80	80-100
Access to employer plan	0.32 (0.01)	0.36 (0.01)	0.34 (0.01)	0.31 (0.01)	0.24 (0.00)
Additional years of access	0.01 (0.00)	0.03 (0.00)	0.04 (0.00)	0.03 (0.00)	0.01 (0.00)
Automatic enrollment	0.28 (0.02)	0.36 (0.01)	0.40 (0.01)	0.27 (0.01)	0.10 (0.01)
Log income	0.00 (0.00)	0.04 (0.00)	0.04 (0.00)	0.04 (0.00)	0.02 (0.00)
Single	-0.04 (0.00)	0.02 (0.00)	0.04 (0.00)	0.03 (0.00)	-0.06 (0.00)
Employer size of Primary filer	-0.03 (0.00)	-0.04 (0.00)	-0.06 (0.00)	-0.05 (0.00)	-0.02 (0.00)
Spouse	-0.06 (0.00)	-0.05 (0.00)	-0.06 (0.00)	-0.04 (0.00)	-0.01 (0.00)
Constant	0.08 (0.00)	0.06 (0.00)	0.17 (0.00)	0.38 (0.00)	0.63 (0.00)
Weak instrument test	2,439	8,453	7,090	4,425	8,165
Observations	5,840,537	3,537,129	1,707,164	791,418	718,509

The three endogenous regressors are an indicator variable for access to an employer retirement plan by either spouse from 2011 to 2019, additional years (beyond the first) that either spouse had access to an employer retirement plan without automatic enrollment, and an indicator variable for automatic enrollment in an employer retirement plan by either spouse. The intent-to-treat instrument is the counterfactual access to an employer retirement plan had the worker remained with the same employer from 2010 to 2019. Robust standard errors, clustered by ZCTA, are reported in parentheses. All specifications include indicator variables for the employer's two-digit NAICS code for the primary filer and spouse (if married), whose coefficients are not reported for brevity. The critical value for a test of weak instruments at the 5 percent significance level is 7.03 (Stock and Yogo, 2005, table 5.2). The sample includes all households with a member aged 50 to 59 in the 2019 administrative tax data, who did not have access to an employer retirement plan in 2010.

Table 7: IV estimates

6.5 Tables 8 and 9: state programs and universal-access counterfactual

Read:

- Table 8 lists state-sponsored retirement savings programs, including start years, default contribution rates, and employer coverage rules.
- These programs motivate the policy counterfactual but are too recent to evaluate in the long run.
- Table 9 predicts how retirement participation would change if universal access with automatic enrollment had already been in effect.
- The predicted increase is 19 percentage points in the lowest income quintile and 16 percentage points in the second quintile.
- The predicted effect is smaller in higher-income groups because baseline access and participation are already high.

Economic message: Universal access with automatic enrollment would especially benefit low- and middle-income households, but it would not achieve universal participation because not all households are working and take-up is incomplete.

Table 8

State-sponsored retirement savings programs.

State	Program	Effective year	Default contribution rate (%)	Employers with at least	
				Workers	Years in business
California	CalSavers	2020	5	5	
Colorado	SecureSavings	2023	5	5	2
Connecticut	MyCTSavings	2022	3	5	
Illinois	Secure Choice	2018	5	5	2
Maine	MaineSaves	2023	5	5	2
Maryland	MarylandSaves	2022	5	1	2
New Jersey	Secure Choice	2022	3	25	2
New York	Secure Choice	2023	3	10	2
Oregon	OregonSaves	2017	5	5	
Virginia	VirginiaSaves	2023	5	25	2

This table reports features of the state-sponsored retirement savings programs as of December 2022.

Table 8: state-sponsored programs

Table 9

Predicted change in retirement account participation.

Participation rate	Percentile of usual income				
	0–20	20–40	40–60	60–80	80–100
Actual	0.21	0.54	0.81	0.93	0.97
Counterfactual	0.40	0.70	0.88	0.96	0.98
Predicted change	0.19	0.16	0.07	0.02	0.01
Observations	7,214,614	7,214,482	7,214,243	7,214,417	7,214,423

In the counterfactual, workers have access to an employer retirement plan with automatic enrollment during all working years. The sample includes all households with a member aged 50 to 59 in the 2019 administrative tax data.

Table 9: predicted participation changes

6.6 Appendix tables: measurement and first-stage evidence

Read:

- Table A.1 compares population counts in tax data and the census. The ratios are close to one, supporting the population coverage of the administrative data.
- Table A.2 reports an alternative bank-account measure that is lower than the benchmark, especially in the lowest income quintile.
- Tables C.1–C.3 report first-stage regressions for the IV analysis.
- The first-stage tables show that the counterfactual employer access instruments strongly predict actual access, additional years, and automatic enrollment.

Economic message: The appendices support both the measurement strategy and the IV design. Population coverage is strong, bank-account measurement is checked, and the instruments have substantial first-stage power.

Table A.1

Comparison of the population count.

Year	Tax data	Census	Ratio
2015	44,221	43,985	1.01
2016	44,202	43,738	1.01
2017	43,920	43,278	1.01
2018	43,463	42,777	1.02
2019	43,332	42,355	1.02

The population counts are reported in thousands. The population count in the administrative tax data includes all individuals aged 50 to 59, who have either a tax return or an information return with a ZIP Code within the U.S. states or Washington, DC. The census population count is the resident population aged 50 to 59.

Table A.1: population count comparison

Table C.1

First-stage regression for access to an employer retirement plan.

Regressor	Percentile of usual income				
	0-20	20-40	40-60	60-80	80-100
Instrument for					
Access to employer plan	0.02 (0.00)	0.16 (0.00)	0.25 (0.00)	0.28 (0.00)	0.42 (0.01)
Additional years of access	0.05 (0.00)	0.02 (0.00)	0.01 (0.00)	0.00 (0.00)	0.01 (0.00)
Automatic enrollment	0.23 (0.00)	0.06 (0.00)	0.01 (0.00)	0.00 (0.00)	0.02 (0.00)
Log income	0.09 (0.00)	0.08 (0.00)	0.01 (0.00)	-0.02 (0.00)	-0.02 (0.00)
Single	-0.02 (0.00)	0.12 (0.00)	0.06 (0.00)	-0.01 (0.00)	-0.11 (0.00)
Employer size of					
Primary filer	0.00 (0.00)	0.04 (0.00)	0.04 (0.00)	0.04 (0.00)	0.01 (0.00)
Spouse	0.03 (0.00)	0.04 (0.00)	0.02 (0.00)	0.02 (0.00)	-0.01 (0.00)
Constant	0.20 (0.00)	0.43 (0.00)	0.61 (0.00)	0.63 (0.00)	1.35 (0.00)
Observations	5,840,537	3,537,129	1,707,164	791,418	718,509

The endogenous regressor is an indicator variable for access to an employer retirement plan by either spouse from 2011 to 2019. The intent-to-treat instrument is the counterfactual access to an employer retirement plan had the worker remained with the same employer from 2010 to 2019. The coefficients for log income and employer size are standardized. Robust standard errors, clustered by ZCTA, are reported in parentheses. All specifications include indicator variables for the employer's two-digit NAICS code for the primary filer and spouse (if married), whose coefficients are not reported for brevity. The sample includes all households with a member aged 50 to 59 in the 2019 administrative tax data, who did not have access to an employer retirement plan in 2010.

Table C.1: first stage for access

Table C.3

First-stage regression for automatic enrollment in an employer retirement plan.

Regressor	Percentile of usual income				
	0-20	20-40	40-60	60-80	80-100
Instrument for					
Access to employer plan	-0.02 (0.00)	-0.01 (0.00)	-0.01 (0.00)	-0.01 (0.00)	-0.01 (0.00)
Additional years of access	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)
Automatic enrollment	0.18 (0.00)	0.27 (0.00)	0.36 (0.00)	0.41 (0.00)	0.54 (0.00)
Log income	0.01 (0.00)	0.03 (0.00)	0.01 (0.00)	0.00 (0.00)	-0.03 (0.00)
Single	0.00 (0.00)	0.04 (0.00)	0.03 (0.00)	0.00 (0.00)	-0.04 (0.00)
Employer size of					
Primary filer	0.00 (0.00)	0.02 (0.00)	0.04 (0.00)	0.04 (0.00)	0.02 (0.00)
Spouse	0.01 (0.00)	0.01 (0.00)	0.02 (0.00)	0.02 (0.00)	0.01 (0.00)
Constant	0.02 (0.00)	0.07 (0.00)	0.14 (0.00)	0.18 (0.00)	0.17 (0.00)
Observations	5,840,537	3,537,129	1,707,164	791,418	718,509

The endogenous regressor is an indicator variable for automatic enrollment in an employer retirement plan by either spouse from 2011 to 2019. The intent-to-treat instrument is the counterfactual access to an employer retirement plan had the worker remained with the same employer from 2010 to 2019. The coefficients for log income and employer size are standardized. Robust standard errors, clustered by ZCTA, are reported in parentheses. All specifications include indicator variables for the employer's two-digit NAICS code for the primary filer and spouse (if married), whose coefficients are not reported for brevity. The sample includes all households with a member aged 50 to 59 in the 2019 administrative tax data, who did not have access to an employer retirement plan in 2010.

2. box 8a of the 2019 form) is 2S on any of its active R.5. Bank branch access

Table A.2

Alternative measure of bank account participation by income.

Year	Percentile of usual income					All
	0-20	20-40	40-60	60-80	80-100	
2015	0.56	0.78	0.86	0.92	0.95	0.82
2016	0.56	0.78	0.86	0.91	0.95	0.81
2017	0.55	0.78	0.86	0.92	0.95	0.81
2018	0.55	0.78	0.87	0.92	0.96	0.82
2019	0.56	0.80	0.88	0.93	0.96	0.83

This table reports an alternative measure of bank account participation for households with a member aged 50 to 59. We define bank account participation as having an account at the most recent tax filing in the previous ten years.

Table A.2: alternative bank account measure

Table C.2

First-stage regression for additional years of access to an employer retirement plan.

Regressor	Percentile of usual income				
	0-20	20-40	40-60	60-80	80-100
Instrument for					
Access to employer plan	-0.18 (0.00)	0.08 (0.01)	0.36 (0.01)	0.47 (0.01)	0.64 (0.01)
Additional years of access	0.19 (0.00)	0.24 (0.00)	0.26 (0.00)	0.29 (0.00)	0.44 (0.00)
Automatic enrollment	0.48 (0.01)	0.14 (0.01)	-0.25 (0.02)	-0.51 (0.03)	-0.40 (0.03)
Log income	0.12 (0.00)	0.36 (0.00)	0.07 (0.00)	-0.07 (0.00)	-0.11 (0.00)
Single	-0.04 (0.00)	0.49 (0.00)	0.37 (0.01)	0.01 (0.01)	-0.32 (0.01)
Employer size of					
Primary filer	-0.02 (0.00)	0.08 (0.00)	0.10 (0.00)	0.10 (0.01)	-0.05 (0.01)
Spouse	0.03 (0.01)	0.10 (0.01)	0.06 (0.01)	0.01 (0.01)	-0.17 (0.01)
Constant	0.22 (0.00)	0.75 (0.00)	1.56 (0.01)	1.70 (0.01)	1.35 (0.00)
Observations	5,840,537	3,537,129	1,707,164	791,418	718,509

The endogenous regressor is additional years (beyond the first) that either spouse had access to an employer retirement plan without automatic enrollment from 2011 to 2019. The intent-to-treat instrument is the counterfactual access to an employer retirement plan had the worker remained with the same employer from 2010 to 2019. The coefficients for log income and employer size are standardized. Robust standard errors, clustered by ZCTA, are reported in parentheses. All specifications include indicator variables for the employer's two-digit NAICS code for the primary filer and spouse (if married), whose coefficients are not reported for brevity. The sample includes all households with a member aged 50 to 59 in the 2019 administrative tax data, who did not have access to an employer retirement plan in 2010.

Table C.2: first stage for years of access

7 Short summary

- The paper studies financial inclusion for the universe of U.S. households with a member aged 50 to 59 in administrative tax data.
- Retirement account participation is very low among low-income households: about 21 percent in the bottom income quintile in 2019.
- Bank account participation is higher than retirement participation but still lower for low-income households.
- Employer retirement plans dominate retirement account participation among low-income households; IRA-only participation is small.
- ZCTA-level average income, inequality, racial composition, and economic connectedness predict retirement participation among low-income households.
- ZCTA-level income inequality predicts bank account participation, while racial composition is less central for bank accounts than for retirement accounts.
- The paper uses a panel IV design based on counterfactual employer plan access had workers stayed with their 2010 employer.
- Access to an employer retirement plan has a large causal effect on retirement participation.
- Automatic enrollment is especially powerful for the bottom income quintile.
- A universal access policy with automatic enrollment could raise retirement participation by 19 percentage points in the bottom income quintile over ten years.
- The broad policy lesson is that financial inclusion depends on access, defaults, geography, and social environment, not only household income.

8 Conclusion

8.1 Core conclusion

Yogo, Whitten, and Cox show that financial inclusion gaps in the United States are large, geographically uneven, and partly driven by employer-based retirement plan access. In administrative tax data, low-income households have much lower retirement account participation, and this gap is closely tied to whether workers have access to employer retirement plans.

8.2 Economic intuition

Retirement participation is costly because opening and maintaining accounts requires access, information, trust, and effort. Employer retirement plans lower these barriers, especially when automatic enrollment is present. Geography matters because households learn from peers, face different employer opportunities, and live in neighborhoods with different levels of inequality and connectedness.

8.3 Policy takeaway

Universal access to employer retirement plans with automatic enrollment is a promising financial inclusion policy. It would not solve all participation gaps, because not all households are working and take-up is incomplete, but it could significantly increase participation for low- and middle-income households. The paper also suggests that researchers and policymakers should use fine geographic data to target financial inclusion interventions.